# Module II

Linear Algebra II

# Rank of the Matrix:

In a non-zero matrix A of order (m x n), if at least one minor of order r is not zero and every minor (r + 1) is zero, then r is said to be the rank of the matrix A and is denoted by $\rho(A)$

- (i) $\rho(A) \geq 0$
- (ii) If A is a matrix of order m x n , then $\rho(A) \leq$ minimum of {m,n}
- (iii) The rank of a zero matrix is '0'
- (iv) The rank of a non- singular matrix of order n x n is 'n'
- (v) The rank of a singular matrix of order n x n is n-1, if minor of any one row is not equal to zero

## Properties of Rank of a Matrix

The properties of the rank of a matrix are as follows:

- Rank ≤ min(m, n) for an m×n matrix
- The rank of a matrix is equal to the number of non-zero rows if it is in Echelon Form.
- Rank(A) = Rank($A^T$), Rank of a matrix and it's transpose are equal.
- The rank of the identity matrix is equal to the order of the identity matrix.
- [Rank of matrix] < [Order of matrix], if it is a singular matrix.
- [Rank of matrix] < [minimum {m, n}], if it is a rectangular matrix of order m x n.
- The rank of a zero matrix or a null matrix is zero.

Find the rank of a matrix $\begin{bmatrix} -1 & -2 & -3 \\ -4 & -5 & -6 \\ -7 & -8 & -9 \end{bmatrix}$ using the Echelon form method.

$$A = \begin{bmatrix} -1 & -2 & -3 \\ -4 & -5 & -6 \\ -7 & -8 & -9 \end{bmatrix}$$

$R_1 \to -R_1$
$R_2 \to -R_2$
$R_3 \to -R_3$

$$= \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$R_2 \to R_2 - 4R_1$
$R_3 \to R_3 - 7R_1$

$$= \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix} \quad R_2 \to \frac{1}{3}R_2,\ R_3 \to \frac{1}{6}R_3$$

$$= \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \quad R_3 \to R_3 - R_2$$

$$= \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad \therefore f(A) = 2$$

: Find the rank of a matrix $\begin{bmatrix} 2 & 4 & 2 & 4 \\ 2 & 6 & 4 & 4 \\ 4 & 8 & 6 & 8 \\ 6 & 14 & 8 & 12 \end{bmatrix}$

$$A = \begin{bmatrix} 2 & 4 & 2 & 4 \\ 2 & 6 & 4 & 4 \\ 4 & 8 & 6 & 8 \\ 6 & 14 & 8 & 12 \end{bmatrix}$$

$R_1 \to \frac{1}{2} R_1$

$R_2 \to \frac{1}{2} R_2$

$R_3 \to \frac{1}{2} R_3$

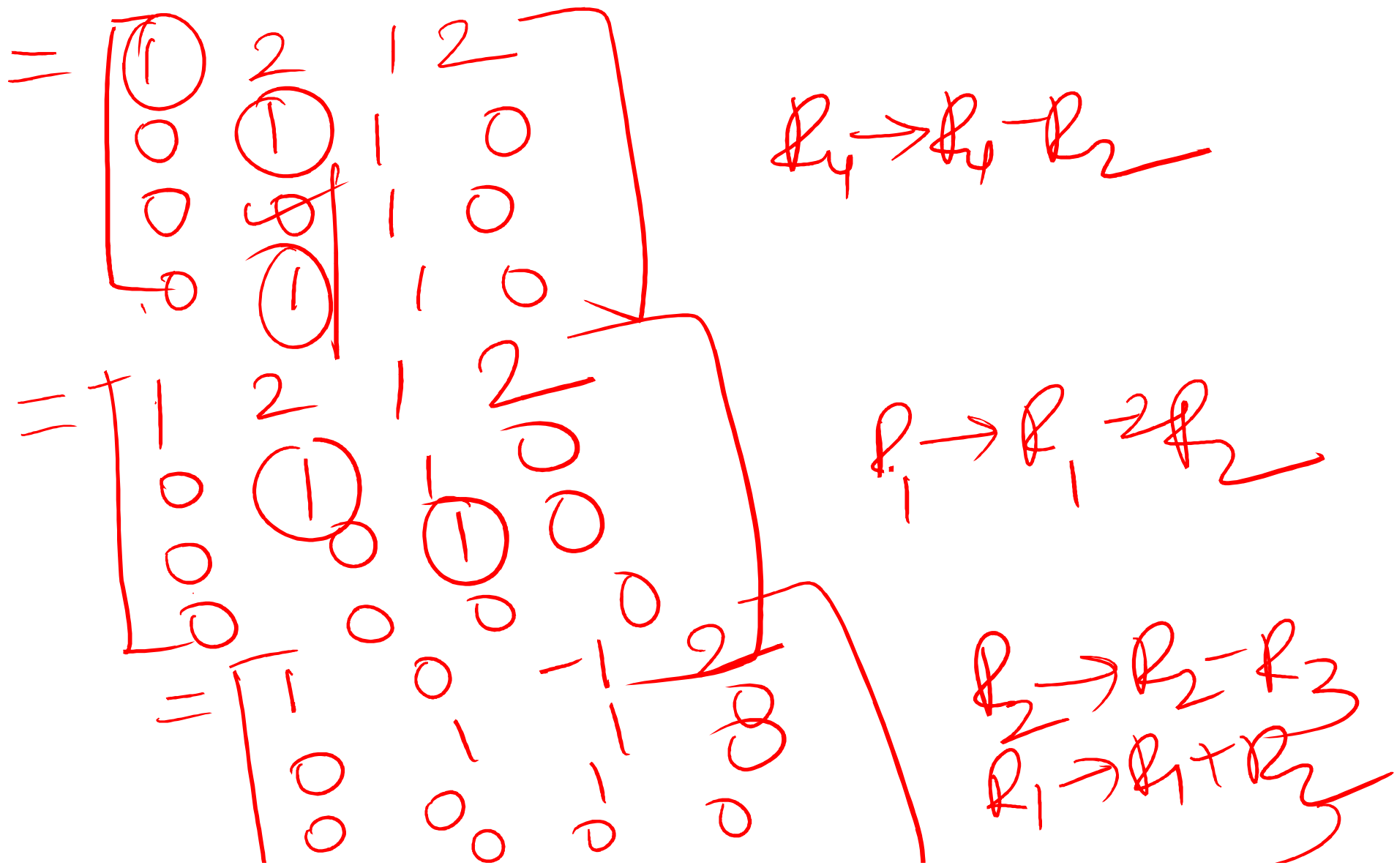
$R_4 \to \frac{1}{2} R_4$

$$= \begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{bmatrix}$$

$R_2 \to R_2 - R_1$

$R_3 \to R_3 - 2R_1$

$R_4 \to R_4 - 3R_1$


$R_4 \to R_4 - R_2$
$R_1 \to R_1 - 2R_2$
$R_2 \to R_2 - R_3$
$R_1 \to R_1 + R_3$

$$= \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore f(A) = 3$

**Reduce the matrix to row echelon form and find the rank**

$$\begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \end{bmatrix}$$

**Reduce the following matrix to echelon form. Find rank**

$$A = \begin{bmatrix} 1 & -1 & 3 & 6 \\ 1 & 3 & -3 & -4 \\ 5 & 3 & 3 & 11 \end{bmatrix}$$

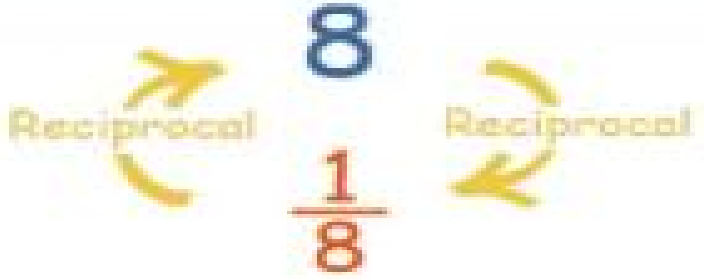
Find the rank of $A = \begin{bmatrix} 2 & 1 & 3 & 5 \\ 4 & 2 & 1 & 3 \\ 8 & 4 & 7 & 13 \\ 8 & 4 & -3 & -1 \end{bmatrix}$.

Determine the rank of by row echelon form.

$$A = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \\ 6 & 13 & 21 & 20 \end{bmatrix}$$

# What is an inverse of a matrix?

- This is the reciprocal of a number.




Reciprocal of a Number

The **Inverse of a Matrix** is the **same idea** but we write it $A^{-1}$



- When we **multiply a number** by its **reciprocal** we get **1**

$$8 \times (1/8) = \mathbf{1}$$

- When we **multiply a matrix** by its **inverse** we get the **Identity Matrix** (which is like "1" for matrices):

$$A \times A^{-1} = \mathbf{I}$$

- Same thing when the inverse comes first:

$$(1/8) \times 8 = \mathbf{1}$$

$$A^{-1} \times A = \mathbf{I}$$

# Inverse of matrix by elementary operations

We can find the inverse of a matrix, if it exists, by using either elementary row operations or column operations but not both

*Note: while using elementary operations, we use*

*Only rows Or ~~Only column~~ Not both*

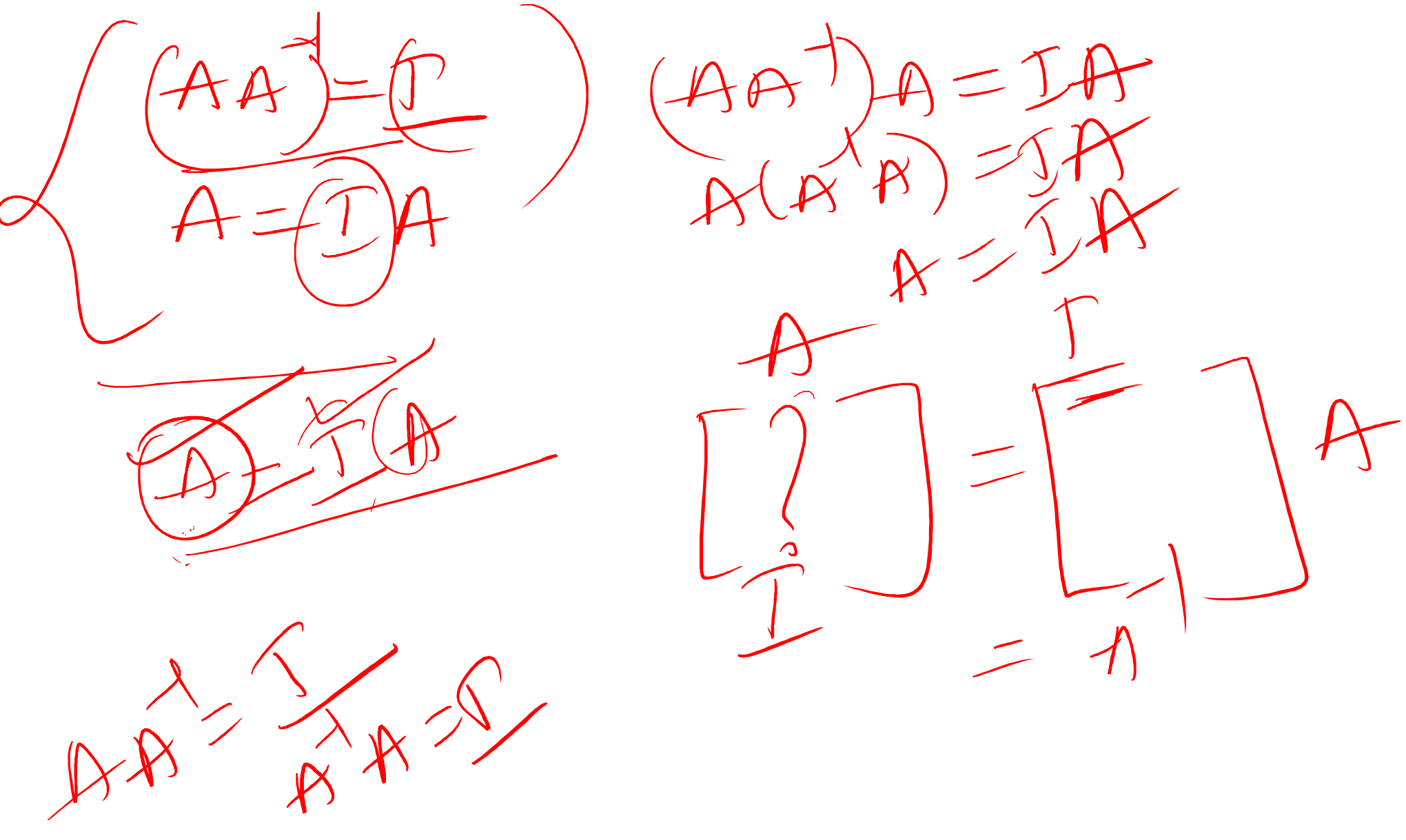
*(i) Let A be an invertible square matrix of order n, the we write* $A = I_n A$.

*(ii) Perform a sequence of elementary row operations on A on the LHS and the pre-factor on the RHS till we obtain the result* $I_n = BA$.

$$A = I_n A$$

**Convert to I using elementary row operations**

**Become** $A^{-1}$



## Inverse Matrix - Definition

- An $n \times n$ matrix $A$ is said to be invertible if there is an $n \times n$ matrix $C$ such that

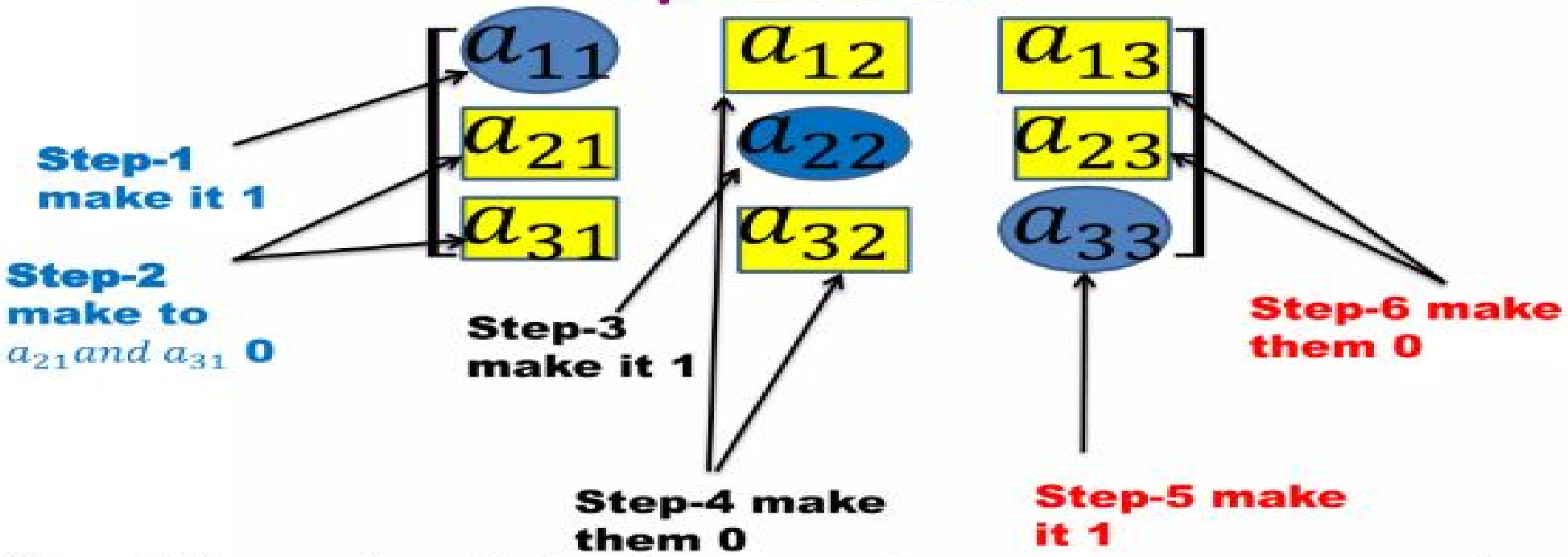
$$CA = I \quad \text{and} \quad AC = I$$

$C = A^{-1}$

where $I = I_n$, the $n \times n$ identity matrix.

- In this case, $C$ is an inverse of $A$.
- This unique inverse is denoted by $A^{-1}$
- $A^{-1}A = I$ and $AA^{-1} = I$
- Invertible = nonsingular
- Not invertible = singular

# Important steps to find inverse of 3x3 order matrix using elementary operations




If we follows these following steps then we can get answer is very short ways.

# Note:

- The inverse of A is $A^{-1}$ only when:

$$A \times A^{-1} = A^{-1} \times A = \mathbf{I}$$

- It should be a square matrix.

- Not all square matrices have inverses. A square matrix which has an inverse is called invertible or nonsingular, and a square matrix without an inverse is called noninvertible or singular.

Example: find the inverse of the matrix $A=\begin{bmatrix}1 & 3\\ 2 & 7\end{bmatrix}$ using elementary row operations.

$$A=\begin{bmatrix}1 & 3\\ 2 & 7\end{bmatrix}$$

$$A = I A$$

$$\begin{bmatrix}1 & 3\\ 2 & 7\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} A$$

$$R_2 \to R_2 - 2R_1$$

$$\therefore I = A^{-1}A$$

$$\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0\\ -2 & 1\end{bmatrix} A$$

$$R_1 \to R_1 - 3R_2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix} A$$

$$\left( I = A^{+} A \right)$$

$$\therefore A^{+} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}$$

Example : find the inverse of the matrix $A=\begin{bmatrix} 6 & -3 \\ -2 & 1 \end{bmatrix}$ using elementary row operations.

Example :find the inverse of the matrix $A=\begin{bmatrix}1 & -1 & 0\\ 2 & 5 & 3\\ 0 & 2 & 1\end{bmatrix}$ using elementary row operations. (CBSE-2015)

$$A=\begin{bmatrix}1 & -1 & 0\\ 2 & 5 & 3\\ 0 & 2 & 1\end{bmatrix}$$

$$A = IA$$

$$\begin{bmatrix}1 & -1 & 0\\ 2 & 5 & 3\\ 0 & 2 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} A \qquad R_2 \to R_2 - 2R_1$$

$$\begin{bmatrix}1 & -1 & 0\\ 0 & 7 & 3\\ 0 & 2 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ -2 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} A \qquad R_2 \to R_2 - 3R_3$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_1 \to R_1 + R_2$

$R_3 \to R_3 - 2R_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 & -3 \\ -2 & 1 & -3 \\ 4 & -2 & 7 \end{bmatrix} A$$

$$A^{-1} = \begin{bmatrix} -1 & 1 & -3 \\ -2 & 1 & -3 \\ 4 & -2 & 7 \end{bmatrix}$$

Example :find the inverse of the matrix $A=\begin{bmatrix} 2 & -1 & 3 \\ -5 & 3 & 1 \\ -3 & 2 & 3 \end{bmatrix}$ using elementary row operations. (CBSE-2016)

$$A=\begin{bmatrix} 2 & -1 & 3 \\ -5 & 3 & 1 \\ -3 & 2 & 3 \end{bmatrix}$$

$$A=IA$$

$$\begin{bmatrix} 2 & -1 & 3 \\ -5 & 3 & 1 \\ -3 & 2 & 3 \end{bmatrix}=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}A \qquad R_1 \to \frac{1}{2}R_1$$

$$\begin{bmatrix} 1 & -1/2 & 3/2 \\ -5 & 3 & 1 \\ -3 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_2 \to R_2 + 5R_1$

$R_3 \to R_3 + 3R_1$

$$\begin{bmatrix} 1 & -1/2 & 3/2 \\ 0 & 1/2 & 17/2 \\ 0 & 1/2 & 15/2 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 & 0 \\ 5/2 & 1 & 0 \\ 3/2 & 0 & 1 \end{bmatrix} A$$

$R_2 \to 2R_2$

$3 - 5 = -2$

$$\begin{bmatrix} 1 & -1/2 & 3/2 \\ 0 & 1 & 17 \\ 0 & 1/2 & 15/2 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 & 0 \\ -5 & 2 & 0 \\ 3/2 & 0 & 1 \end{bmatrix} A$$

$R_1 \to R_1 + R_3$

$$= \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 17 \\ 0 & +11/2 & 15/2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 1 \\ +5 & 2 & 0 \\ 3/2 & 0 & 1 \end{bmatrix} A \qquad R_3 \to \cancel{2}R_3$$

$$= \begin{bmatrix} 1 & 0 & 9 \\ & & \end{bmatrix} - \begin{bmatrix} 2 & 0 & 1 \\ & & \\ 3 & 0 & 2 \end{bmatrix} A \qquad R_3 \to R_3 + R_2$$

$$\begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 17 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 1 \\ 5 & 2 & 0 \\ -2 & -2 & 2 \end{bmatrix} \qquad R_3 \to \frac{1}{2}R_3$$

$$\begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 17 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 1 \\ 5 & 2 & 0 \\ 1 & 1 & -1 \end{bmatrix} A$$

$R_1 \to R_1 - 9R_3$

$R_2 \to R_2 - 17R_3$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -7 & -9 & 10 \\ -12 & -15 & 17 \\ 1 & 1 & -1 \end{bmatrix} A$$

$$A^{-1} = \begin{bmatrix} -7 & -9 & 10 \\ -12 & -15 & 17 \\ 1 & 1 & -1 \end{bmatrix} //$$

$$A = \begin{bmatrix} 4 & -2 & 3 \\ 8 & -3 & 5 \\ 7 & -2 & 4 \end{bmatrix}.$$

Find the inverse of the following matrix

$$(1) \quad A = \begin{bmatrix} 1 & 4 & 3 \\ -1 & -2 & 0 \\ 2 & 2 & 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -6 & 2 & 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 & -1 \\ 1 & -1 & 2 \\ 1 & 1 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ -1 & 2 & 1 \end{bmatrix}$$

# Introduction

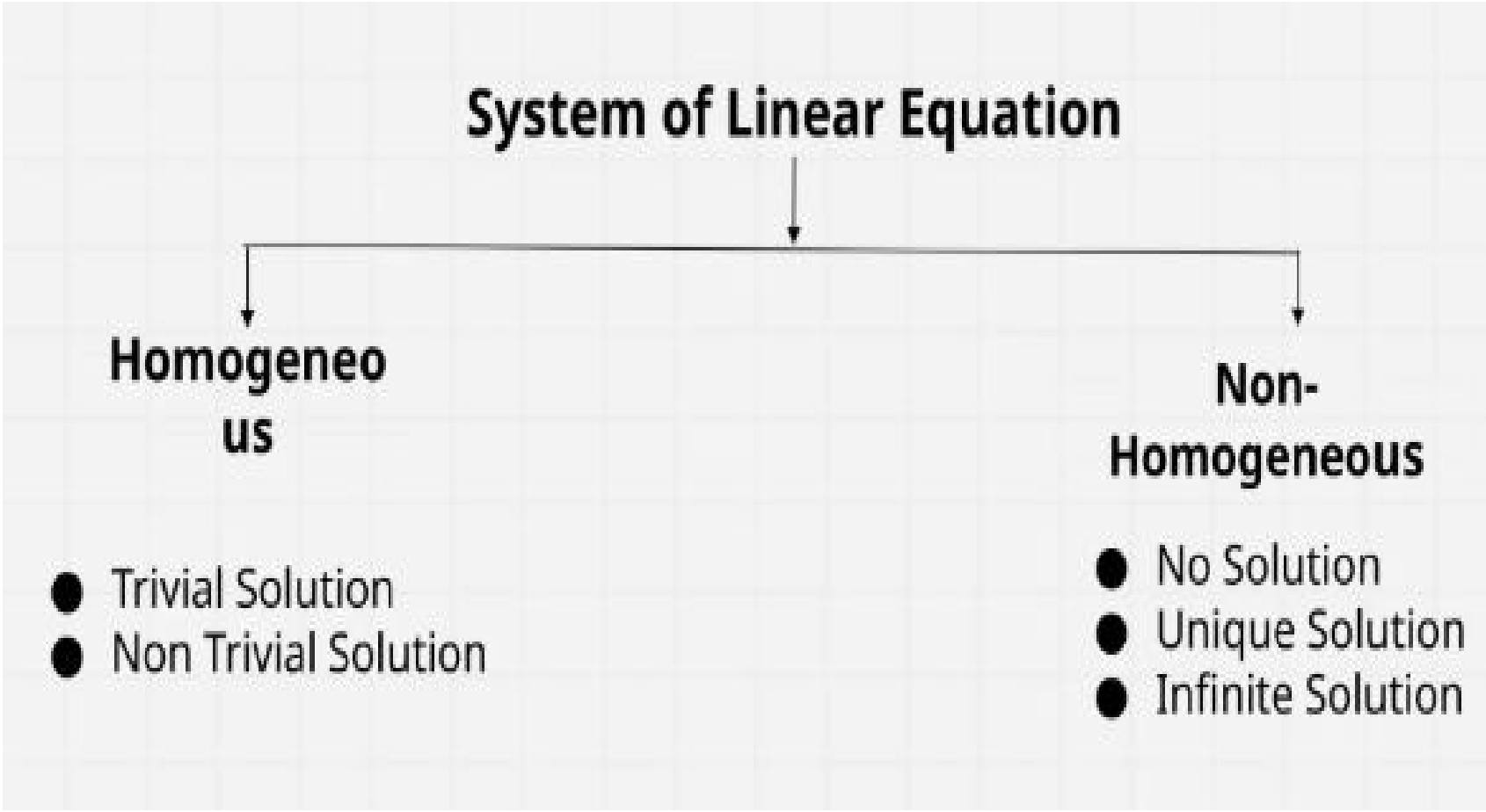
- A system of linear equations consists of two or more linear equations with the same variables.
- General form:

$$a_1x + b_1y + c_1z = d_1,$$

$$a_2x + b_2y + c_2z = d_2,$$

$$a_3x + b_3y + c_3z = d_3$$

- Solution gives values of x, y, z satisfying all equations.


System of Linear Equation
Homogeneous
Trivial Solution
Non Trivial Solution
Non-Homogeneous
No Solution
Unique Solution
Infinite Solution

## 1.1. SYSTEMS OF LINEAR EQUATIONS

- A system of linear equations has

1. no solution, or
2. exactly one solution, or
3. infinitely many solutions.

The linear system is **inconsistent**. ← 1

The linear system is **consistent**. ← 2, 3

Trivial soln

Non-Trivial soln

# Consistent or Inconsistent System?

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 2 \end{array}\right] \qquad \begin{array}{l} x = 1 \\ y = 5 \\ z = 2 \end{array}$$

*One solution: Consistent system*

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 0 & 2 \end{array}\right] \qquad \begin{array}{l} x = 4 \\ y = 7 \\ 0 = 2 \end{array}$$

*No solution: Inconsistent system*

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right] \qquad \begin{array}{l} x + 5z = 2 \\ y + 3z = 2 \\ z = \text{any real number} \end{array}$$

*Infinite solutions: Consistent system*

$$(x = 2 - 5z, y = 2 - 3z, z = \text{any real number})$$

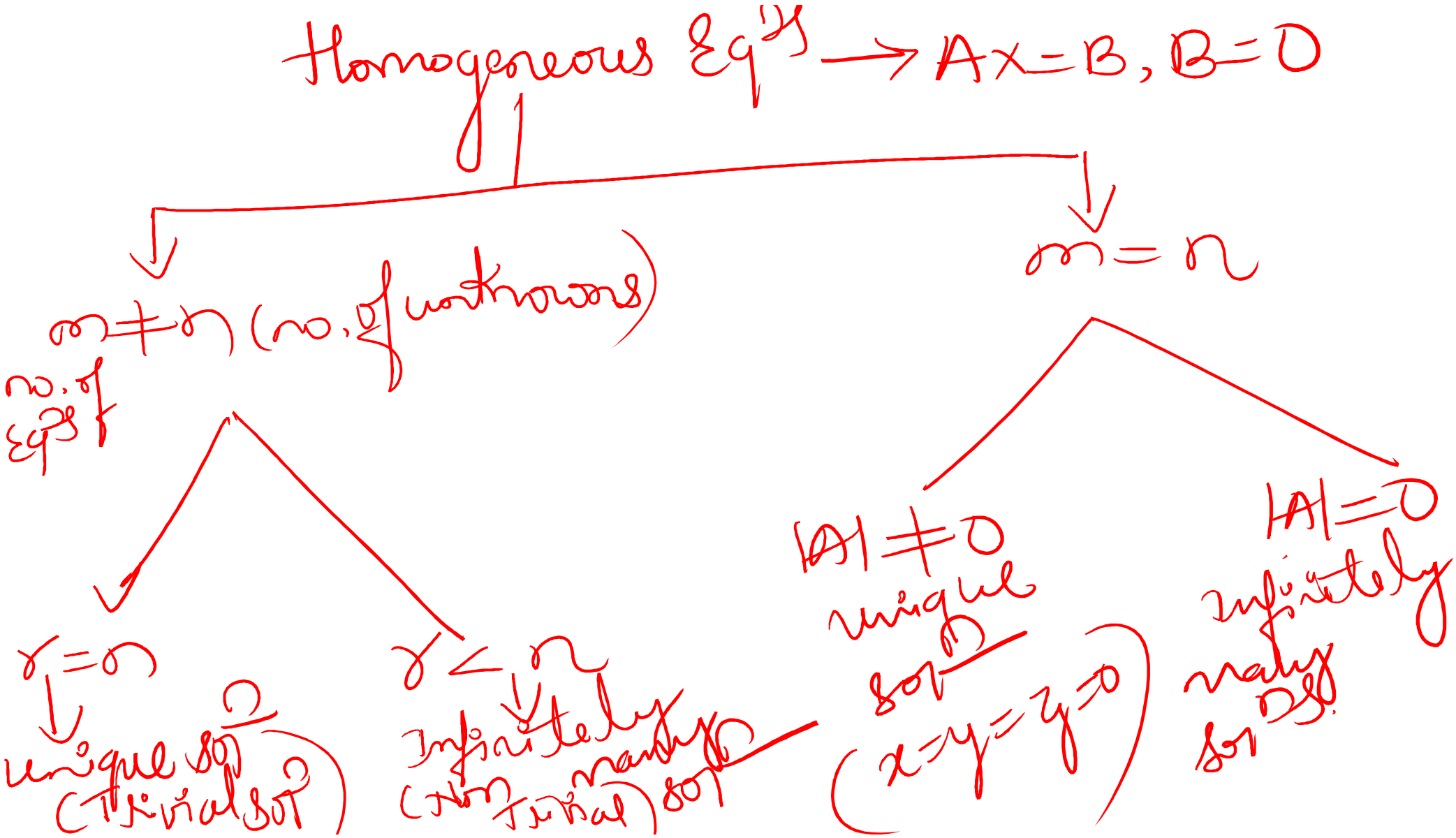
*or*

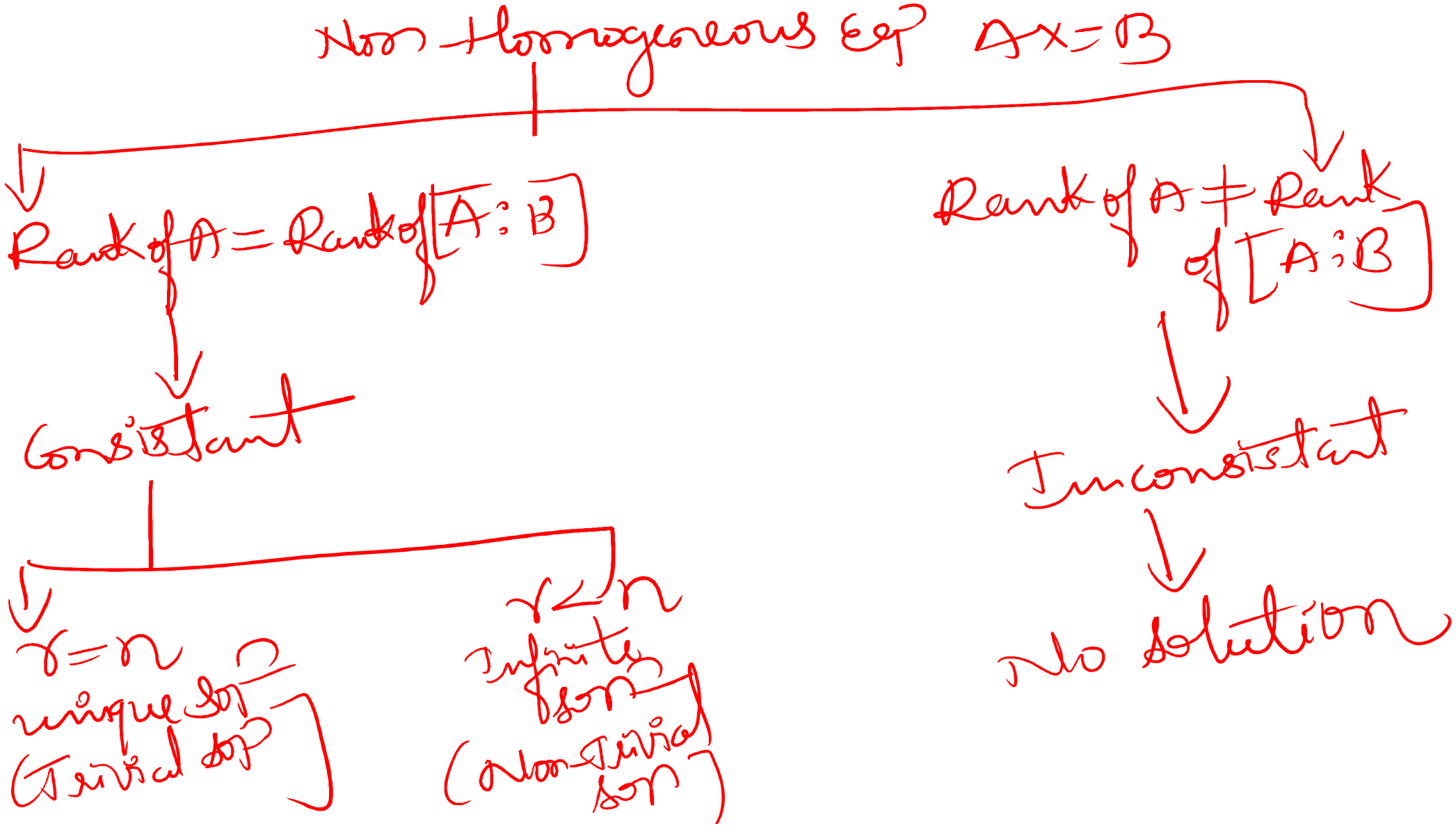
$$\{(x, y, z) | x = 2 - 5z, y = 2 - 3z, z = \text{any real number}\}$$

## Homogeneous systems of linear equations:

A system of linear equations is said to be homogeneous if all the constant terms are zero.

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= 0 \\
a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= 0 \\
a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \cdots + a_{3n}x_n &= 0 \\
&\vdots \\
a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n &= 0
\end{aligned}
$$


Homogeneous Eq^ns → AX=B, B=0
m≠n (no. of unknowns)
m=n
no. of eqns
r=n
unique sol^n (Trivial sol^n)
r<n
Infinitely many sol^ns (non-trivial sol^ns)
|A|≠0
unique sol^n
(x=y=z=0)
|A|=0
infinitely many sol^ns


Non Homogeneous Eq AX=B
Rank of A = Rank of [A:B]
Rank of A ≠ Rank of [A:B]
Consistant
Inconsistant
No solution
r=n
unique sol
(Trivial sol)
r<n
Infinite sol
(non-Trivial sol)

Examine the non-trivial soln of the system of eqns

$$5x+2y-3z=0$$
$$3x+y+z=0$$
$$2x+y+6z=0$$

$m=n$

$$A=\begin{bmatrix} 5 & 2 & -3 \\ 3 & 1 & 1 \\ 2 & 1 & 6 \end{bmatrix}$$

$$|A| = 5(6-1) - 2(18-2) - 3(3-2)$$
$$= 25 - 32 - 3 \neq 0 = -10 \neq 0$$

$\therefore$ System of Eq has unique sol$^{n}$

$$[A:B] = \left[\begin{array}{ccc:c} 5 & 2 & -3 & 0 \\ 3 & 1 & 1 & 0 \\ 2 & 1 & 6 & 0 \end{array}\right] \quad R_1 \to R_1 - 2R_3$$

$$= \left[\begin{array}{ccc:c} 1 & 0 & -15 & 0 \\ 3 & 1 & 1 & 0 \\ 2 & 1 & 6 & 0 \end{array}\right] \quad \begin{matrix} R_2 \to R_2 - 3R_1 \\ R_3 \to R_3 - 2R_1 \end{matrix}$$

$$= \left[\begin{array}{ccc:c} 1 & 0 & -15 & 0 \\ 0 & 1 & 46 & 0 \\ 0 & 1 & 36 & 0 \end{array}\right] \quad R_3 \to R_3 - R_2$$

$$= \begin{bmatrix} 1 & 0 & -15 & : & 0 \\ 0 & 1 & 46 & : & 0 \\ 0 & 0 & -10 & : & 0 \end{bmatrix}$$

$R_3 \to -\frac{1}{10} R_3$

$$= \begin{bmatrix} 1 & 0 & -15 & : & 0 \\ 0 & 1 & 46 & : & 0 \\ 0 & 0 & 1 & : & 0 \end{bmatrix}$$

$R_1 \to R_1 + 15R_3$

$R_2 \to R_2 - 46R_3$

$$= \begin{bmatrix} 1 & 0 & 15 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \end{bmatrix}$$

$x = y = z = 0$

## Homogeneous Linear equations

Solving a system of m equations in n unknowns

1) *Write* $AX = O$

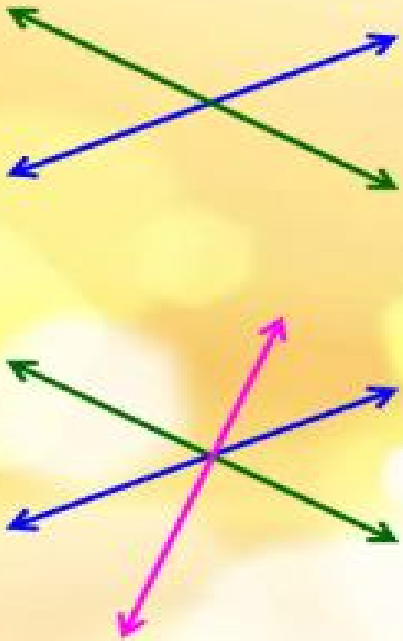
2) *Apply elementary row transformations only and reduce A to echelon form.*

3) *If* $r = n$, *only trivial slotion is posssible.* $(x = 0, y = 0, z = 0)$

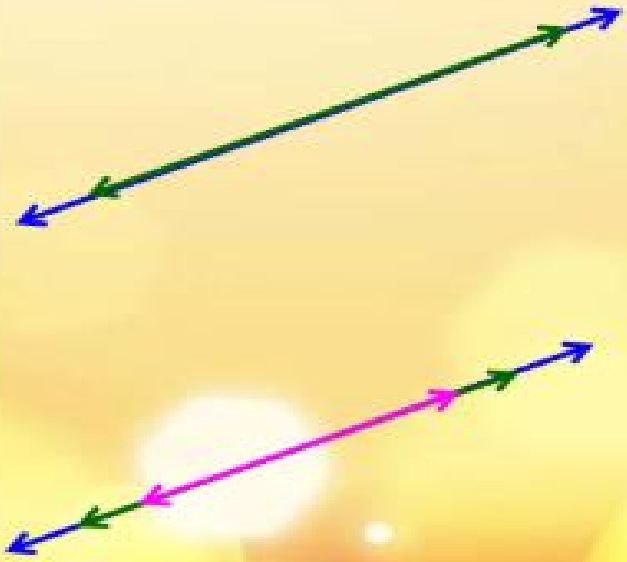
4) *If* $r < n$, *non−trivial solution is posssible.*

# Types of Solutions

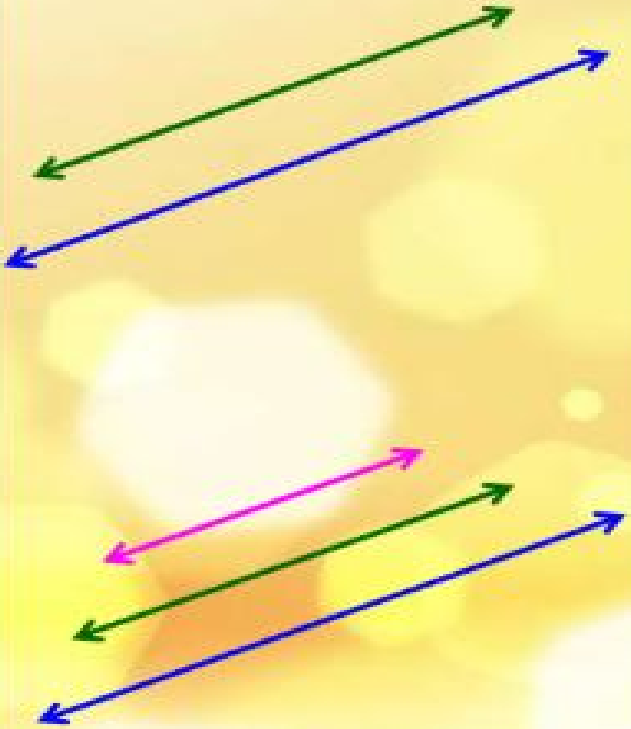
Consistent System
One solution



Consistent System
Infinite solutions



Inconsistent Syste
No solution



$$4x - 3y - 2z = 0$$

$$5x + 9y + 23z = 0$$

$$3x + 2y + 7z = 0$$

$$
\begin{aligned}
x_1 &+ x_2 &+ x_3 &= 0\\
x_1 &+ 2x_2 &+ x_3 &= 0\\
x_1 &+ x_2 &+ 2x_3 &= 0
\end{aligned}
$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

By ref

$R_2 \to R_2 - R_1$

$R_3 \to R_3 - R_1$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\rho(A) = 3 =$ no. of unknowns

$r = 3 = n$

$\therefore$ system has unique soln

$\therefore x_1 = x_2 = x_3 = 0$

$$
\begin{aligned}
x_1 &+ x_2 &+ x_3 &= 0 \\
x_1 &+ 2x_2 &+ x_3 &= 0 \\
2x_1 &+ 3x_2 &+ 2x_3 &= 0
\end{aligned}
$$

1. Solve the following system of homogenous equations.

(i) $3x + 2y + 7z = 0,\ 4x - 3y - 2z = 0,\ 5x + 9y + 23z = 0$

(ii) $2x + 3y - z = 0,\ x - y - 2z = 0,\ 3x + y + 3z = 0$

(i) $A = \begin{bmatrix} 3 & 2 & 7 \\ 4 & -3 & -2 \\ 5 & 9 & 23 \end{bmatrix}$ $R_2 \to R_2 - R_1$

$= \begin{bmatrix} 3 & 2 & 7 \\ 1 & -5 & -9 \\ 5 & 9 & 23 \end{bmatrix}$ $R_2 \leftrightarrow R_1$

$= \begin{bmatrix} 1 & -5 & -9 \\ 3 & 2 & 7 \\ 5 & 9 & 23 \end{bmatrix}$ $R_2 \to R_2 - 3R_1$, $R_3 \to R_3 - 5R_1$

$$= \begin{bmatrix} 1 & -5 & -9 \\ 0 & 17 & 34 \\ 0 & 34 & 68 \end{bmatrix}$$

$R_2 \to \frac{1}{17} R_2$

$R_3 \to \frac{1}{34} R_3$

$$= \begin{bmatrix} 1 & -5 & -9 \\ 0 & 1 & 2 \\ 0 & 4 & 2 \end{bmatrix}$$

$R_3 \to R_3 - R_2$

$$= \begin{bmatrix} 1 & -5 & -9 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore \rho(A) = 2 = r$

$n = 3$

$r \neq n$

$= 3 - 2$

$= 1$

$\therefore$ System has infinitely many sols.

$z = t, \quad t \in \mathbb{R}$

$\therefore$
$$\begin{cases} x - 5y - 9z = 0 \\ y + 2z = 0 \end{cases}$$

$y = -2z$

$y = -2t$

$x = 5y + 9z = 5(-2t) + 9(t) = -10t + 9t$

$$x = -t$$

$$\therefore\ x = -t,\ y = -2t,\ z = t,\ t \in \mathbb{R}$$

Solve $x_1 + x_2 - x_3 + x_4 = 0$

$x_1 - x_2 + 2x_3 - x_4 = 0$

$3x_1 + x_2 + x_4 = 0$

$$A = \begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix} \quad \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - 3R_1 \end{matrix}$$

$$= \begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & -2 & 3 & -2 \\ 0 & -2 & 3 & -2 \end{bmatrix} \quad R_3 \to R_3 - R_2$$

$$= \begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & -2 & 3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\rho(A) = 2$, $n = 4$

$r \neq n$

$\therefore$ System has infinitely many sol^n s.

$x_4 = t_1$, $x_3 = t_2$, $t_1, t_2 \in \mathbb{R}$ [$\because n - r = 4 - 2 = 2$

$$x_1 + x_2 - x_3 + x_4 = 0$$

$$2x_2 + 3x_3 - 2x_4 = 0$$

$$-2x_2 = 2x_4 - 3x_3$$

$$x_2 = \frac{3}{2}x_3 - x_4$$

$$x_2 = \frac{3t_2}{2} - t_1$$

$$x_1 = -x_2 + x_3 - x_4$$

$$= -\left(\frac{3}{2}t_2 - t_1\right) + t_2 - t_1$$

$$= t_1 - \frac{3}{2}t_2 + t_2 - t_1$$

$$= -\frac{t_2}{2}$$

$$x_1 = -\frac{t_2}{2},\ x_2 = \left(\frac{3}{2}t_2 - t_1\right),\ x_3 = t_2$$

$$x_4 = -t_1$$

Does the system have a nontrivial solution? if yes determine

1. $4x - y + 2z + t = 0, \quad 2x + 3y - z - 2t = 0,$
   $7y - 4z - 5t = 0, \qquad 2x - 11y + 7z + 8t = 0$

**Ans:** $x = -\frac{1}{4}(5k + s), y = \frac{1}{7}(4k + 5s), z = k, t = s$

**Example1**: *Is the system have non trivial solution? If it has find.*

$3x + 4y - z - 6w = 0; \quad 2x - 3y + 2z - 3w = 0;$

$2x + y - 14z - 9w = 0; \quad x + 3y + 13z + 3w = 0.$

Test the consistency of the system

$x+2y+z=3$, $2x+3y+2z=5$, $3x-5y+5z=2$,

$3x+9y-z=4$ and solve if it is consistent.

$$[A:B] = \left[\begin{array}{ccc:c} 1 & 2 & 1 & 3 \\ 2 & 3 & 2 & 5 \\ 3 & -5 & 5 & 2 \\ 3 & 9 & -1 & 4 \end{array}\right]$$

$R_2 \to R_2 - 2R_1$

$R_3 \to R_3 - 3R_1$

$R_4 \to R_4 - 3R_1$

$$= \left[\begin{array}{ccc:c} 1 & 2 & 1 & 3 \\ 0 & -1 & 0 & -1 \\ 0 & -11 & 2 & -7 \\ 0 & 3 & -4 & -5 \end{array}\right]$$

$R_3 \to R_3 - 11R_2$

$R_4 \to R_4 + 3R_2$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & -4 & -8 \end{array}\right] \qquad R_3 \to \frac{1}{2}R_3, \quad R_4 \to -\frac{1}{4}R_4$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & \textcircled{1} & 2 \\ 0 & 0 & 1 & 2 \end{array}\right] \qquad R_4 \to R_4 - R_3$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\therefore \rho(A:B) = 3 = \rho(A)$

$\therefore$ System is consistent and It has unique sol$^n$

$\therefore$ $z = 2$, $-y = -1$, $x + 2y + z = 3$

$y = 1$ $\quad x + 2(1) + 2 = 3$

$\therefore x+4=3,\ x=-1$

$\therefore x=-1, y=1, z=2$

Test the consistency and solve if consistent for the system $3x-y+4z=3$, $x+2y-3z=-2$, $6x+5y-5z=-3$.

$$[A:B] = \begin{bmatrix} 3 & -1 & 4 & : & 3 \\ 1 & 2 & -3 & : & -2 \\ 6 & 5 & -5 & : & -3 \end{bmatrix} \quad R_2 \leftrightarrow R_1$$

$$= \begin{bmatrix} 1 & 2 & -3 & : & -2 \\ 3 & -1 & 4 & : & 3 \\ 6 & 5 & -5 & : & -3 \end{bmatrix} \quad \begin{matrix} R_3 \to R_3 - 6R_1 \\ R_2 \to R_2 - 3R_1 \end{matrix}$$

$$= \left[\begin{array}{ccc:c} 1 & 2 & -3 & -2 \\ 0 & -7 & 13 & 9 \\ 0 & -7 & 13 & 9 \end{array}\right] \quad R_3 \to R_3 - R_2$$

$$= \left[\begin{array}{ccc:c} 1 & 2 & -3 & -2 \\ 0 & -7 & 13 & 9 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\therefore \rho(A:B) = 2 = \rho(A)$

$\therefore$ System is consistant or it has unique solution

$$\begin{cases} 2x + y - z = 3 \\ x - 2y + 4z = -1 \\ 3x + y + 2z = 7 \end{cases}$$

$$-2x+y-z=4,\quad x+2y+3z=13,\quad 3x+z=-1$$

# Eigen Vector:

- In linear algebra , an eigenvector or characteristic vector of a square matrix is a vector that does not changes its direction under the associated linear transformation.

- In other words – If V is a vector that is not zero, than it is an eigenvector of a square matrix A if Av is a scalar multiple of v. This condition should be written as the equation:

$$AV = \lambda v$$

# Eigen Value:

- In above equation λ is a scalar known as the **eigenvalue** or **characteristic value** associated with eigenvector **v.**

- We can find the eigenvalues by determining the roots of the characteristic equation-

$$|A-\lambda I|=0$$

① Find Eigen Value and Eigen Vector of the matrix $A = \begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix}$

$A - \lambda I = \begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$

$= \begin{bmatrix} -4-\lambda & 2 \\ 4 & 3-\lambda \end{bmatrix}$

$|A-\lambda I|=0, \begin{vmatrix} -4-\lambda & 2 \\ 4 & 3-\lambda \end{vmatrix}=0$ $x^2+(a+b)x+ab=0$

$= (-4-\lambda)(3-\lambda) - 2\times 4 = 0$

$= -12+4\lambda-3\lambda+\lambda^2-8 = 0$

$= \lambda^2+\lambda-20 = 0$ $\quad -20$

$= \lambda^2+5\lambda-4\lambda-20=0$ $\quad 5 \quad -4$

$\lambda(\lambda+5)-4(\lambda+5)=0$

$(\lambda+5)(\lambda-4)=0$

$\lambda=-5, 4$

$\therefore$ Eigen Values $\lambda = -5, 4$

Let $\lambda_1 = -5$

$$(A - \lambda_1 I)x = 0$$

$$\begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix} - (-5)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\left\{ \begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix} + \begin{bmatrix} +5 & 0 \\ 0 & +5 \end{bmatrix} \right\} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \qquad R_2 \to \frac{1}{4}R_2$$

$$\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \qquad R_2 \to R_2 - R_1$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$x_1 + 2x_2 = 0$

Take $x_2 = t$, $x_1 + 2(t) = 0$

$x_1 = -2t$, $x_1 = -2t$

Eigen Vector

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2t \\ t \end{bmatrix} = t\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\lambda_2 = 4$$

$$(A - \lambda_2 I)x = 0$$

$$\left( \begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix} - 4 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) x = 0$$

$$\begin{bmatrix} -8 & 2 \\ 4 & -1 \end{bmatrix} x = 0 \qquad R_1 \to \frac{1}{2} R_1$$

$$\begin{bmatrix} -4 & 1 \\ +4 & -1 \end{bmatrix} x = 0 \qquad R_2 \to R_2 + R_1$$

$$\begin{bmatrix} -4 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \neq 0 \\ x_2 \end{bmatrix}$$

$$-4x_1 + x_2 = 0$$

Take $x_2 = t$, $x_1 = \frac{t}{4}$

or $\begin{bmatrix} x_1 = t, & x_2 = 4t \end{bmatrix}$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} t \\ 4t \end{bmatrix} = t \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

③ Find Eigen Value and Eigen Vector of the the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix} = 0$$

$$\left[ \because \lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0 \right]$$

$S_1 = tr(A)$

$S_2 =$ sum of all principle diagonal minors

$$|A| = \begin{vmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{vmatrix} = 8(21-16) + 6(-18+8) + 2(24-14)$$

$$= 8(5) + 6(-10) + 2(10)$$

$$= 40 - 60 + 20 = 0$$

$$|A| = 0$$

$$\text{tr}(A) = 8+7+3 = 18$$

$$S_2 = \begin{vmatrix} 7 & -4 \\ -4 & 3 \end{vmatrix} + \begin{vmatrix} 8 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix} = 45$$

$S_1$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$$

$$\lambda^3 - 18\lambda^2 + 45\lambda = 0$$

$$\lambda(\lambda^2 - 18\lambda + 45) = 0$$

$$\lambda = 0, \quad \lambda = +15, +3$$

45

$-15$ $-3$

$\lambda = 0, 3, 15$ are Eigen Values., $\lambda_1 = 0$

$$[A - \lambda I]X = [A - 0]X = AX = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} =$$

$$8x_1 - 6x_2 + 2x_3 = 0$$
$$-6x_1 + 7x_2 - 4x_3 = 0$$
$$2x_1 - 4x_2 + 3x_3 = 0$$

By Cramer's Rule

$$\frac{x_1}{\begin{vmatrix} -6 & 2 \\ 7 & -4 \end{vmatrix}} = \frac{-x_2}{\begin{vmatrix} 8 & 2 \\ -6 & -4 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix}} = t$$

$$\frac{x_1}{10} = \frac{-x_2}{-20} = \frac{x_3}{20} = t$$

$$\frac{x_1}{1} = \frac{x_2}{2} = \frac{x_3}{2} = t$$

$$x_1 = t$$
$$x_2 = 2t$$
$$x_3 = 2t$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$\lambda_2 = 3$$

$$[A - \lambda_2 I]X = 0$$

$$[A - 3I]X = 0$$

$$\left\{ \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \right\} X = 0$$

$$\begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$5x_1 - 6x_2 + 2x_3 = 0$$

$$-6x_1 + 4x_2 - 4x_3 = 0$$

$$2x_1 - 4x_2 + 0 = 0$$

$$A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}$$

$$\lambda^3-3\lambda^2-9\lambda-5=0$$

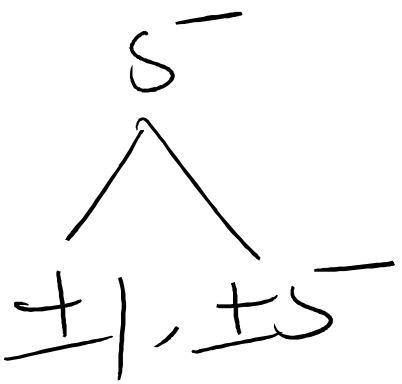
$$\lambda(\lambda^2-3\lambda-9)-5=0$$

$5$

$\pm 1, \pm 5$

$$\lambda^3-S_1\lambda^2+S_2\lambda-|A|=0$$

| | | | | |
|---|---|---|---|---|
| $-1$ | $1$ | $-3$ | $-9$ | $-5$ |
| | $0$ | $-1$ | $4$ | $5$ |
| | $1$ | $-4$ | $-5$ | $0$ |




$$(\lambda+1)(\lambda^2-4\lambda-5)=0$$

$$(\lambda+1)=0 \quad (\lambda^2-4\lambda-5)=0$$

$$\lambda=-1, \quad (\lambda^2-5\lambda+\lambda-5)=0$$

$$\lambda=-1. \qquad \lambda=-1,5$$



$\therefore$ Eigen values are $\lambda = -1, -1, 5$

$\lambda_1 = -1$

$$[A - \lambda I]X = 0$$

$$[A + I]X = 0$$

$$\left\{\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\right\} X = 0$$

$$\begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_2 \to R_2 - R_1$

$R_3 \to R_3 - R_1$

$$\begin{bmatrix} 2 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$R_1 \to \frac{1}{2} R_1$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$x_1 + x_2 + x_3 = 0$

~~$x_2 = t$, $x_3 = 0$~~ or $x_2 = 0$, $x_3 = s$

or

~~$x_2 = t$, $x_3 = s$~~

$x_1 = -t$

$x_2 = t$ $X = \begin{bmatrix} -t \\ t \\ 0 \end{bmatrix}$, $X = t\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

$x_3 = 0$

$x_1 = -s$

$x_2 = 0$

$x_3 = s$

$X = \begin{bmatrix} -s \\ 0 \\ s \end{bmatrix} = s\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

$\therefore$ Eigen vectors $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ & $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ for $\lambda = -1$

$\lambda_2 = 5$

∴ Synthetic division method

$\pm 1$

$\pm 5$

| | | | | |
|---|---|---|---|---|
| $-1$ | $1$ | $-3$ | $-4$ | $-5$ |
| | $0$ | $-1$ | $+4$ | $5$ |
| | $1$ | $-4$ | $5$ | $0$ |

$(\lambda+1)(\lambda^2-4\lambda-5)=0$

$-5$

$-5+1$

$(\lambda+1)(\lambda+1)(\lambda+1)(\lambda-5)=0$

$-1,-1,5$

Find the eigenvalue and eigenvector of matrix A.

$$A=\begin{bmatrix} -4 & 1 & 0 \\ 0 & -3 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

2. $\begin{bmatrix} 5 & -2 \\ 9 & -6 \end{bmatrix}$

3. $\begin{bmatrix} 3 & 5 & 3 \\ 0 & 4 & 6 \\ 0 & 0 & 1 \end{bmatrix}$

4. $\begin{bmatrix} 13 & 5 & 2 \\ 2 & 7 & -8 \\ 5 & 4 & 7 \end{bmatrix}$